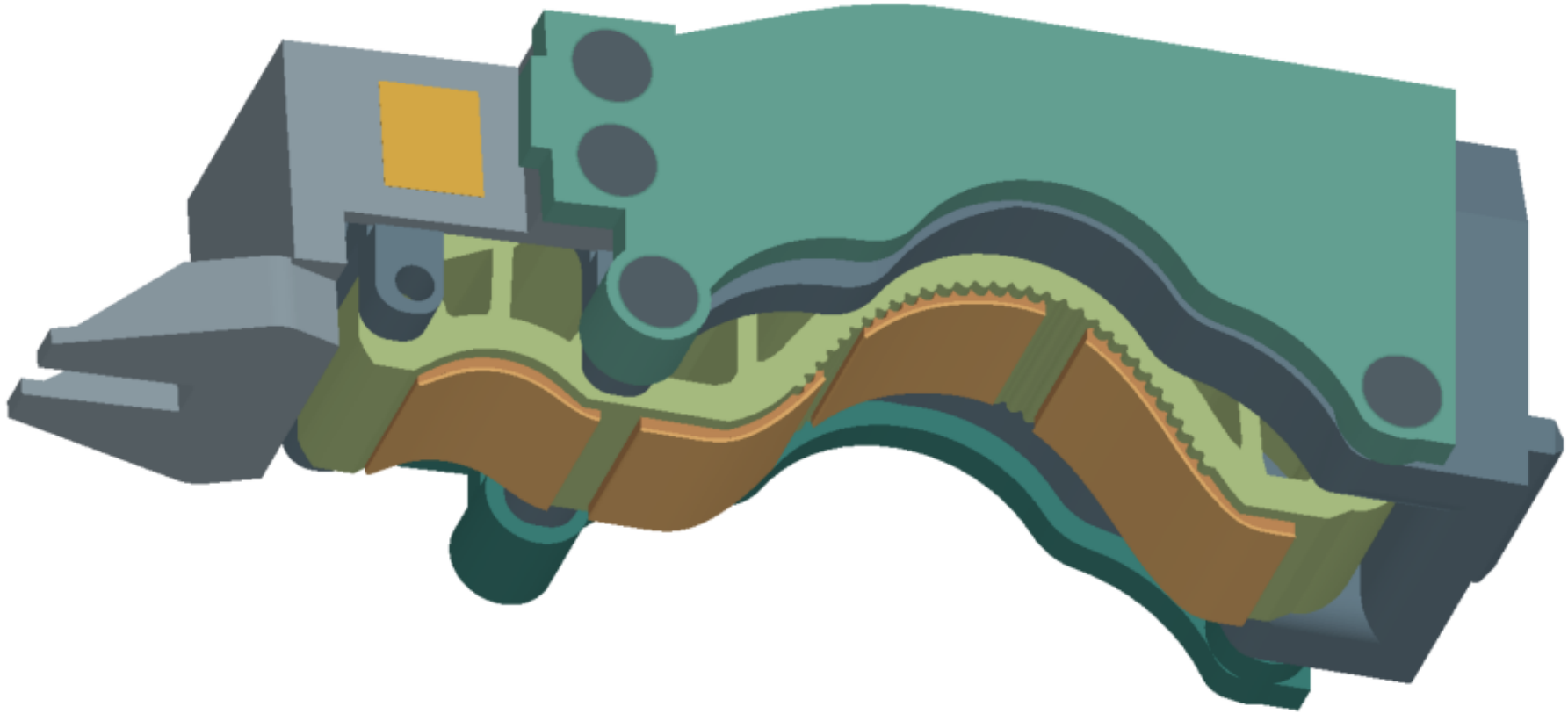
N1 · bare sensing nail with four TPU contact cells

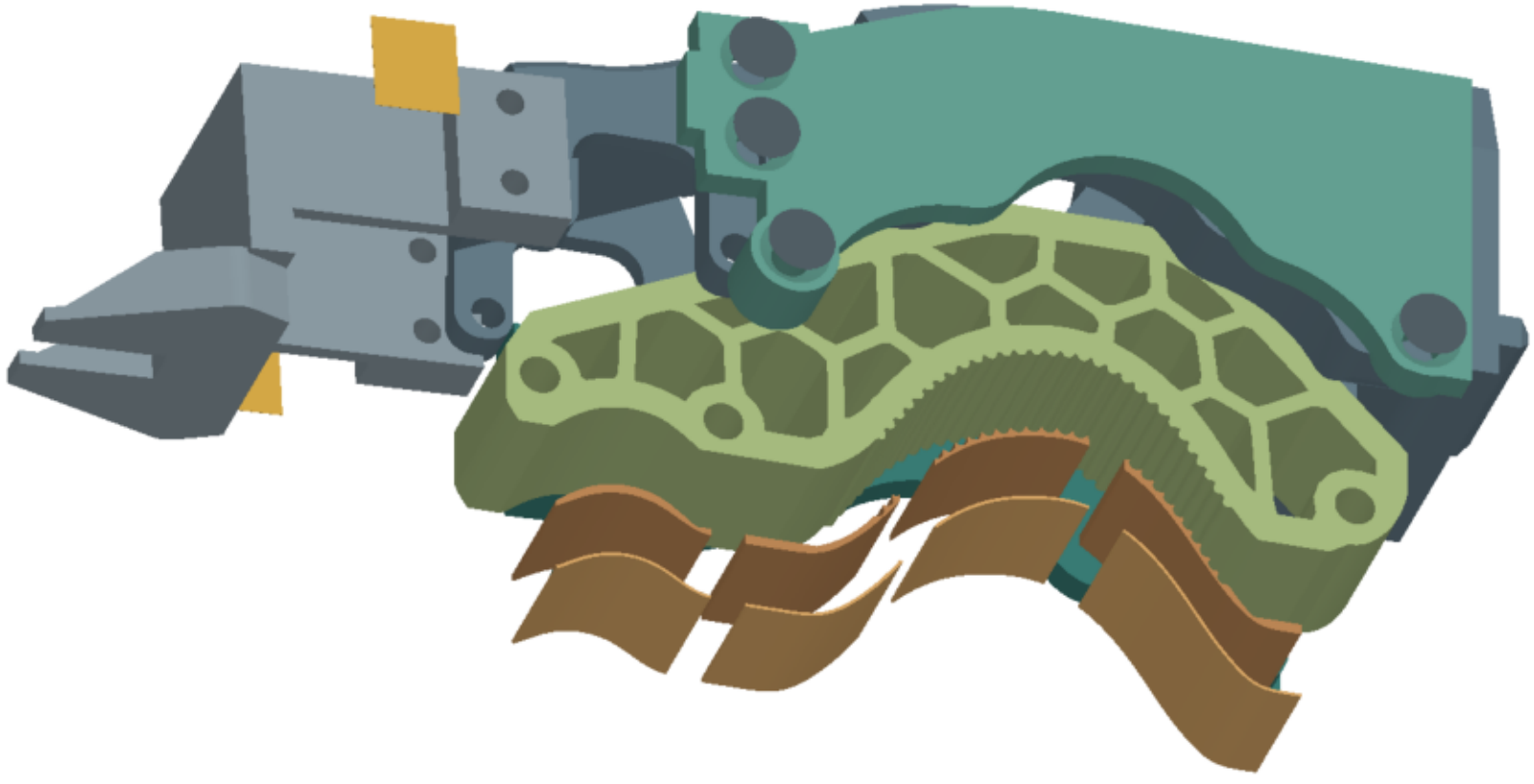


Steel fork + two shear webs · fixed side carriers · revised PETG frame · original TPU + four R1 cells

CAD rendering. Gold rectangles locate purchased foil carriers; wiring, adhesive and protective coatings are not modeled.

Experimental prototype · physical assembly, calibration and qualification pending

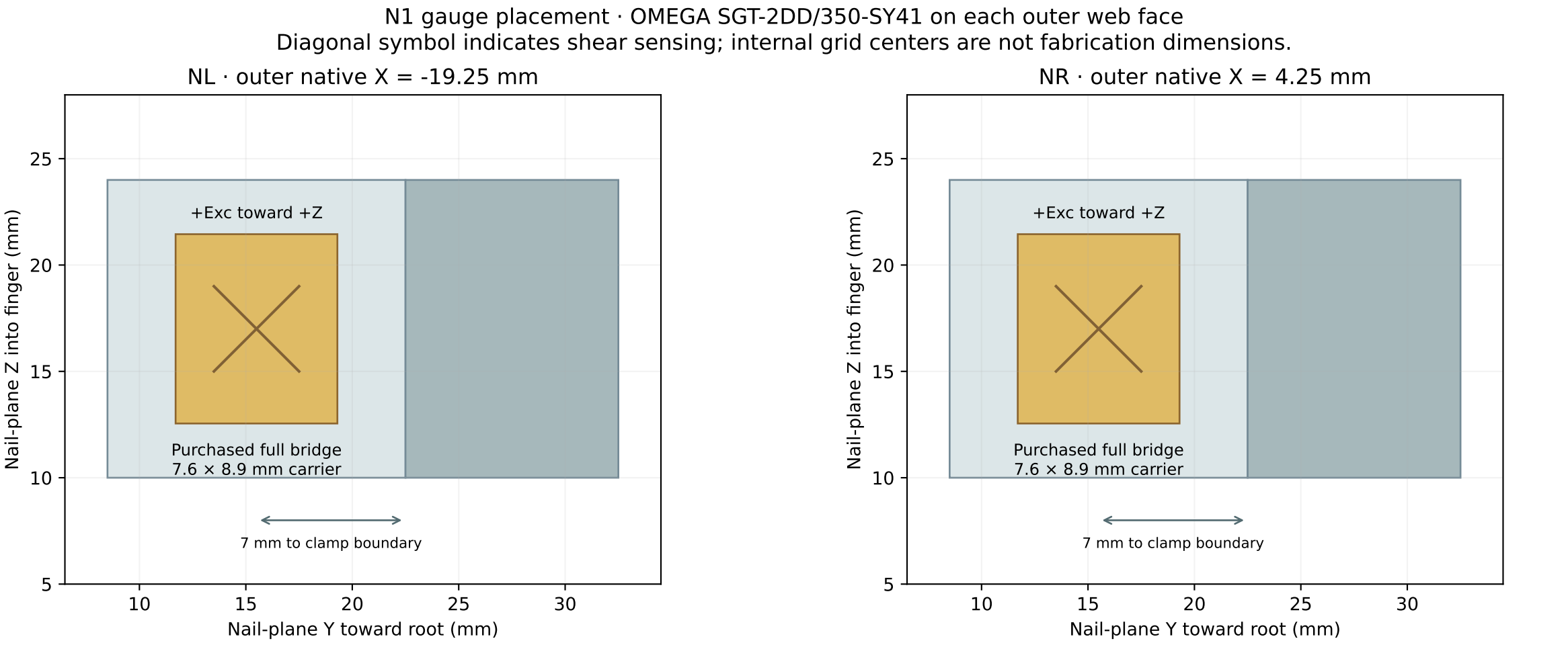
N1 · separate assembly components



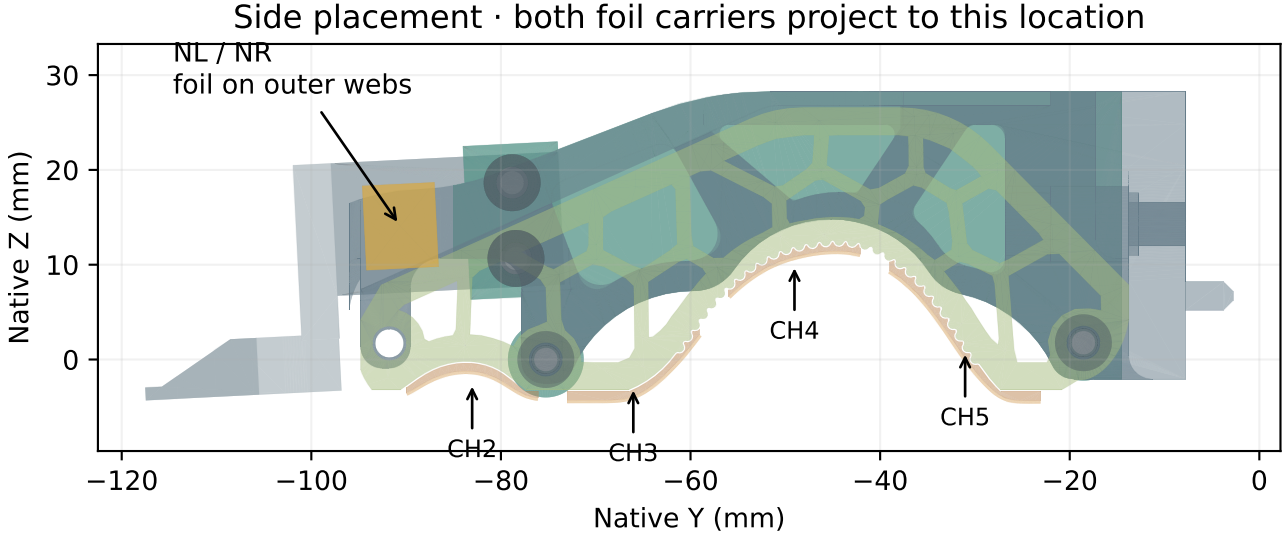
Steel fork + two shear webs · fixed side carriers · revised PETG frame · original TPU + four R1 cells

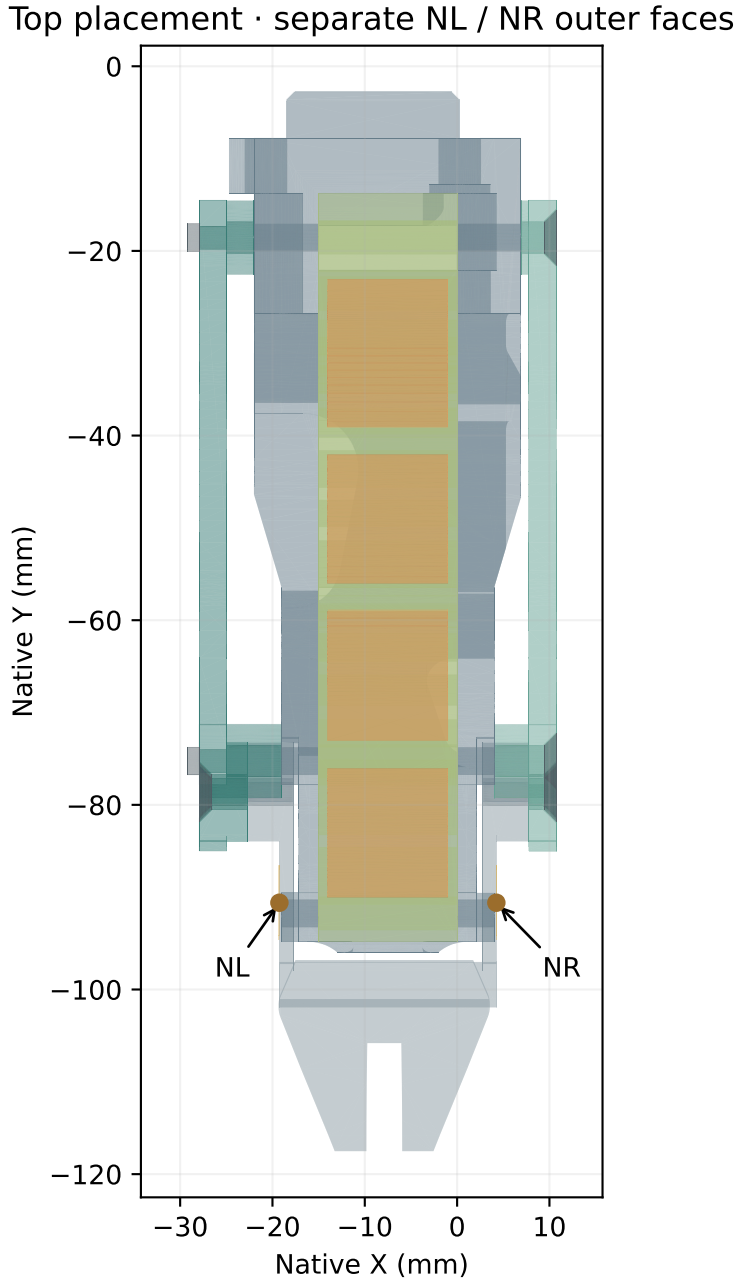
CAD rendering. Gold rectangles locate purchased foil carriers; wiring, adhesive and protective coatings are not modeled.

Experimental prototype · physical assembly, calibration and qualification pending


N1 gauge placement · OMEGA SGT-2DD/350-SY41 on each outer web face
Diagonal symbol indicates shear sensing; internal grid centers are not fabrication dimensions.
NL · outer native X = -19.25 mm
NR · outer native X = 4.25 mm
+Exc toward +Z
Purchased full bridge
7.6 × 8.9 mm carrier
7 mm to clamp boundary
Nail-plane Z into finger (mm)
Nail-plane Y toward root (mm)
5
10
15
20
25
30

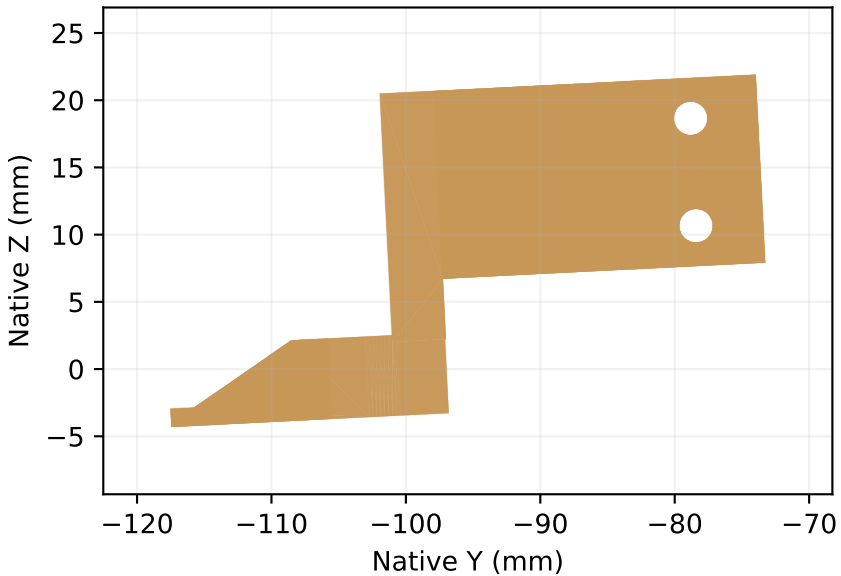
N1 sensing locations · two nail bridges and four retained R1 TPU cells

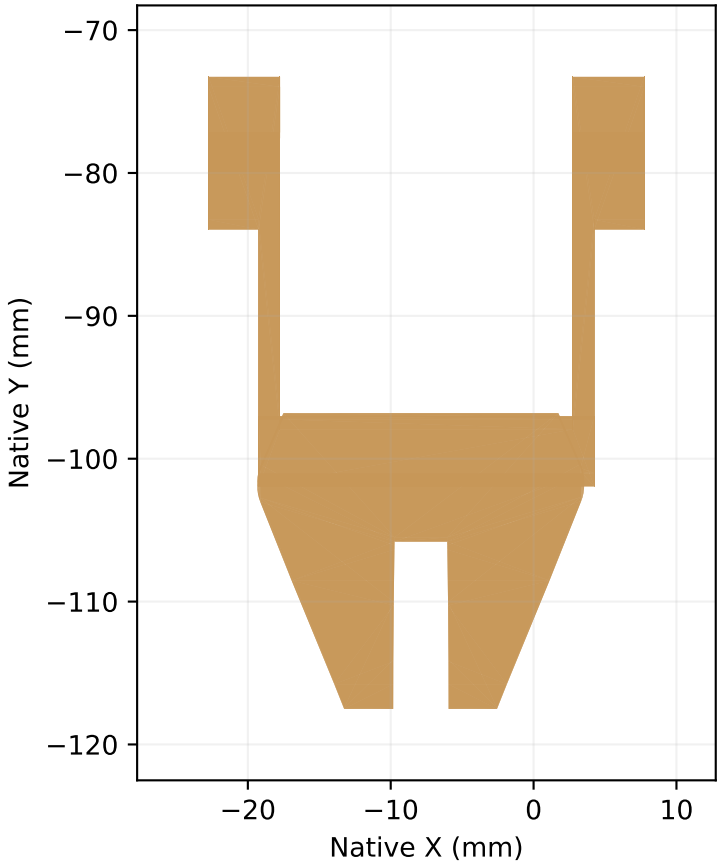


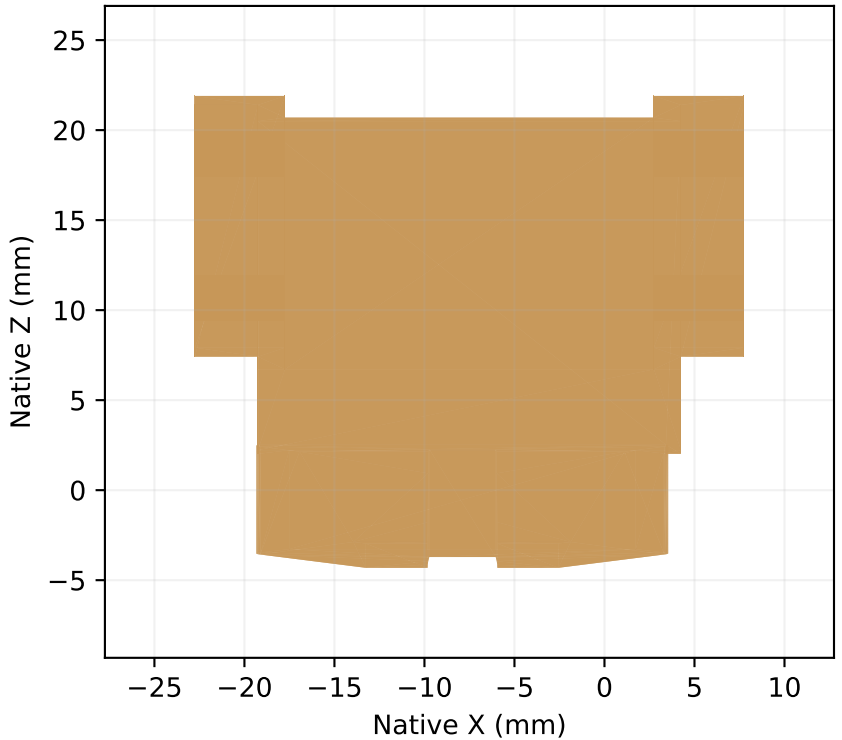


nail_bare_sensing_fork

Bare source-shaped leading fork, two vertical shear webs and thick inactive clamp ends.

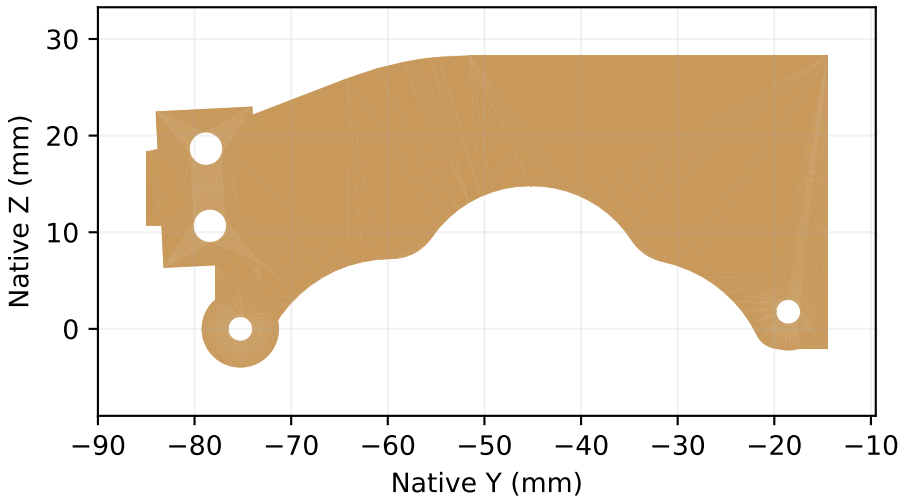


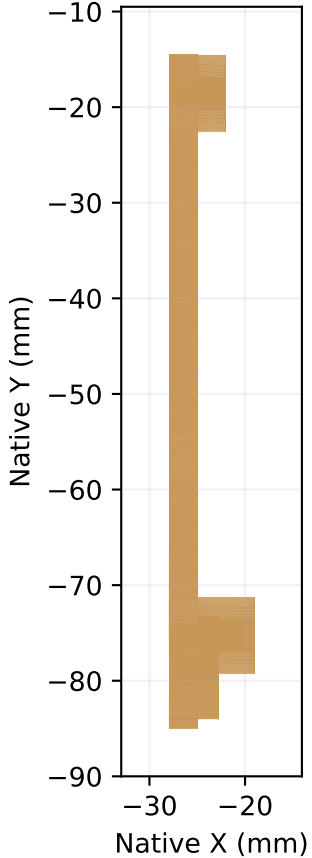


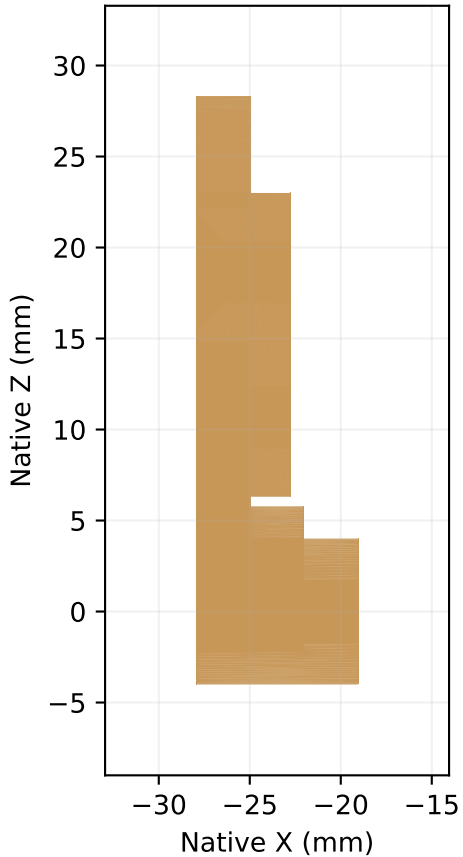


nail_left_fixed_carrier

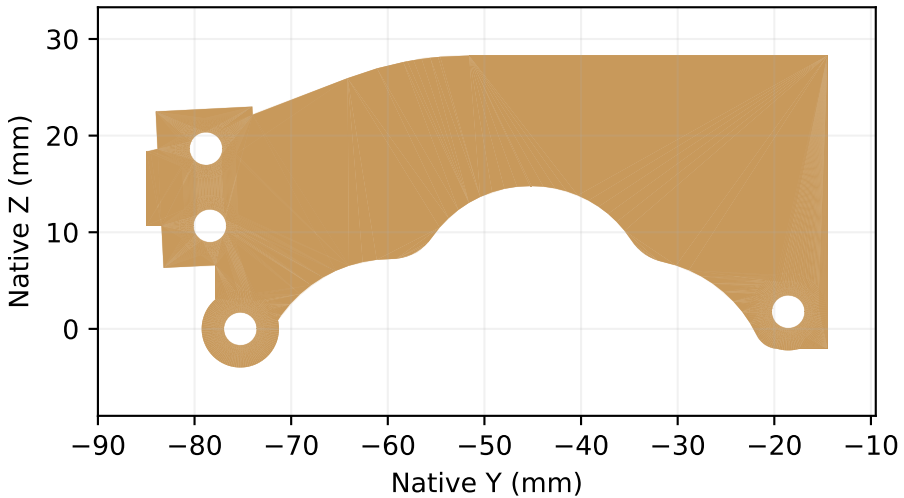
Fixed metal support, original side attachment axes, M3 threaded mount and extended front clamp shoe.

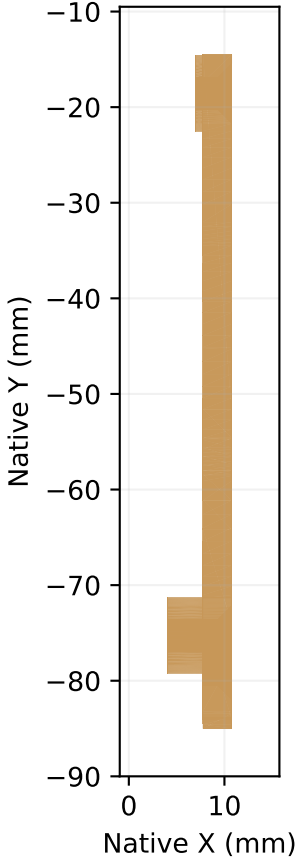


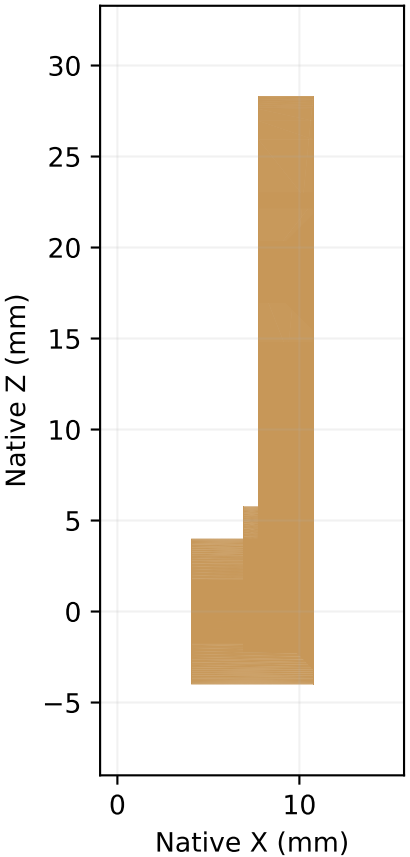




nail_right_fixed_carrier
Fixed metal support with flush M3 side-mount and front-clamp screws.

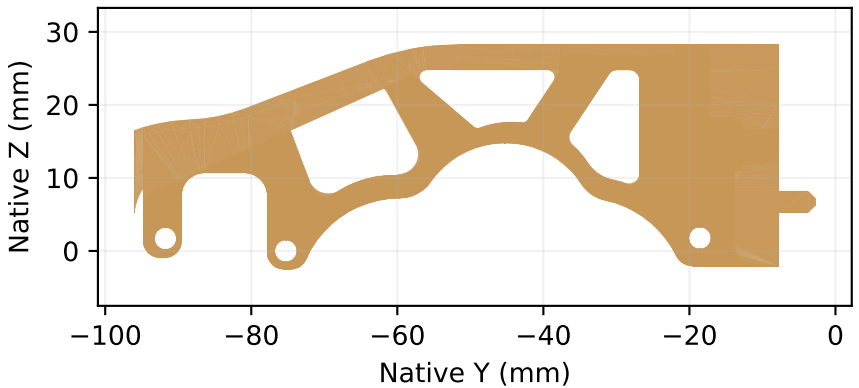


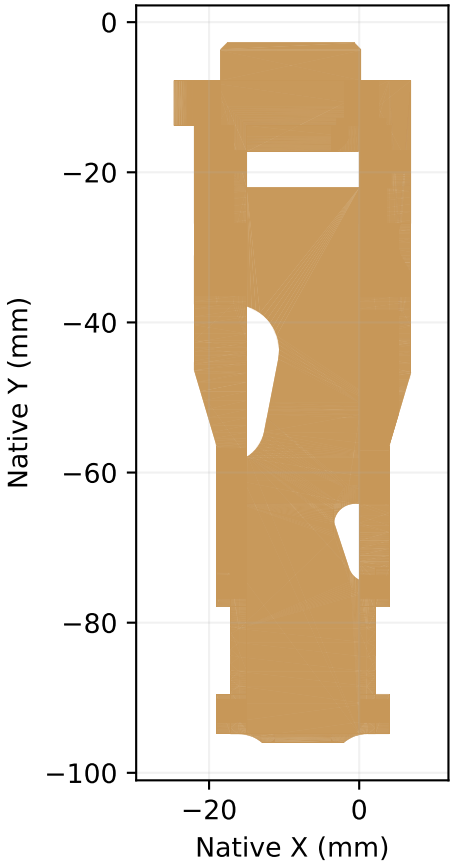


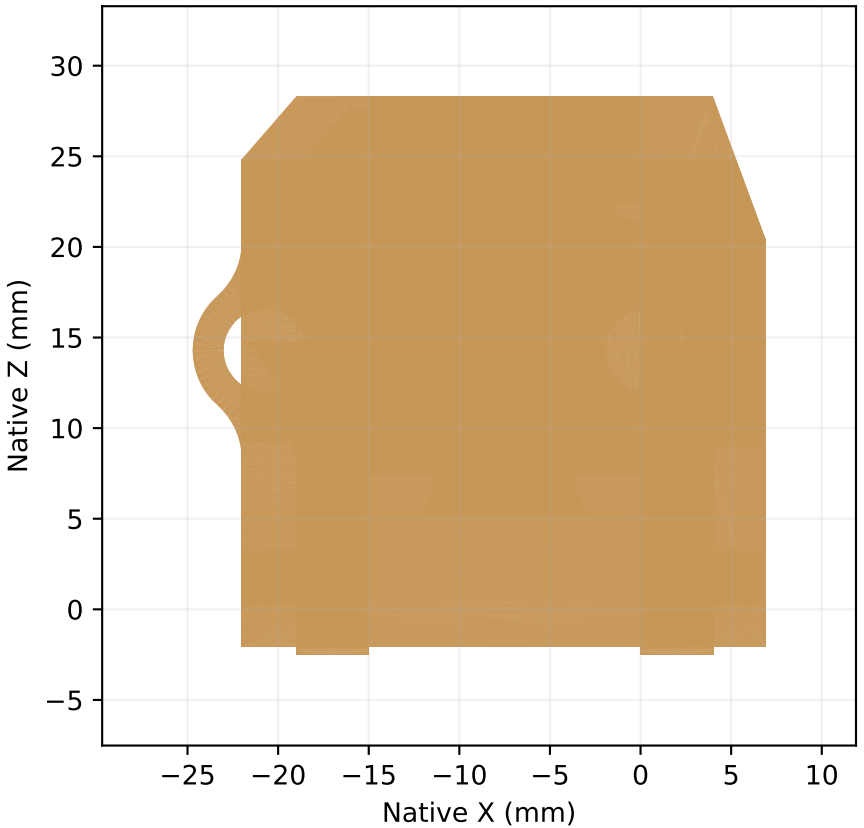


nail_revised_hard_frame

Separate original-frame derivative with its hard nail removed; source root and TPU attachment geometry retained.







N1.1 prototype review · millimeters · use STEP masters

Nail: wrought 17-4PH, H900 candidate; certify the actual material.
Leading contact-plane footprint retained; leading thickness 1.35 mm.
Two webs: 1.5 mm thick × 14 mm tall; active length 14 mm.
Each foil carrier center: nail-plane Y=15.5, Z=17 mm.
Full shear bridge: OMEGA SGT-2DD/350-SY41, 350 ohm.

Front clamps: two M3x8 flat heads left; two M3x6 right.
Side mounting: M3x40 flat heads from right into tapped left carrier.
Inspect actual screw engagement, source posts and clearances.
Tap-drill CAD contains 2.5 mm bores; cut M3 threads as specified.

Four TPU cells use R1 CH2–CH5 tooling and material instructions.
Their 0.30 mm laminate envelopes are displayed; covers are not.
Use the complete six-file N1 firmware and its raw recorder.

The frame cut's small distal crescent is intentionally removed.
The three attachment regions and central frame are checked.
Clearance pockets are in the new frame; original files are retained.

No physical build, force calibration, contact reliability or
robot-task performance has been measured for this design.